EXIT TICKET / REFLECTION

## Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**HTTPS guarantees the website operator is honest.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

### GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.





DO NOW — RECALL

## Lesson 132 · Hybrid Encryption: Protecting a Connection

Name: _____ Class: _____ Date: _____

**1. Which key verifies a signature produced using an owner's private key?**

**2. Which memory is typically volatile: RAM or ROM?**

**3. In a queue, what does FIFO stand for?**

**4. What is O(n) search complexity associated with?**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

The corresponding trusted public key.

### 2. Model answer

RAM.

### 3. Model answer

First in, first out.

### 4. Model answer

Linear/sequential search in the worst case.

### By the end, you can…

- Client validates server identity and securely establishes shared key material.
- Both ends obtain secret symmetric traffic keys.
- Bulk application data travels encrypted under symmetric cipher.

### Vocabulary — know these terms

session key — Temporary symmetric key for a connection/session.

hybrid encryption — Combines asymmetric key establishment with symmetric data encryption.

certificate — Credential binding identity to a public key, subject to trust validation.

confidentiality — Preventing unauthorised reading of information.



RHS COMPUTING OCR H446 · LESSON 132 · DRAFT

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Evaluate the assertion 'the padlock means any entered data are safe'. [8]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Describe why hybrid cryptography uses asymmetric and symmetric methods together. [6]

### Question 2

Explain what a session key is and why it is not the same as a website's public key. [6]





NEW LEARNING

# Read, annotate and explain

### Authenticate and agree

Client validates server identity and securely establishes shared key material.

### Derive session key

Both ends obtain secret symmetric traffic keys.

### Encrypt traffic

Bulk application data travels encrypted under symmetric cipher.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

**Explain why a modern web connection does not encrypt every file byte directly with an asymmetric cipher.**

### Check server certificate

Predict the first step from the prompt.

### Derive symmetric traffic key

Explain the intermediate result or decision.

### Encrypt application data

Modern secure connections typically authenticate with public-key credentials, establish shared session keys (often via ephemeral key agreement) and use efficient symmetric encryption for bulk traffic. Identity checks guard against impostors.





WE DO — GUIDED PRACTICE

# Try → Discuss → Improve